

EE 553: Foundations of Intelligent Communication Systems-I

Course Code	EE 553
Course Name	Foundations of Intelligent Communication Systems-I
L-T-P-C	3-0-2-4
Prerequisites	IC 111 Linear Algebra, IC 260 or equivalent
Intended for	MTech/MTech (R)/ PhD students of the SCEE
Mutual Exclusion	EE522, CS512, MA512, MA525, MA551, or equivalent
Approval	54th BoA (May 2024)

Course Contents

- **Signal Space Representation:** Lowpass representation of bandpass signals, vector spaces, signal space concepts, subspaces, linear dependence/ independence, basis and dimension, sampling and reconstruction, inner product, orthogonal expansion of signals, orthogonal matrices, four fundamental subspaces. (9 Hours)
- **Linear Time-Invariant Systems:** Response, properties, eigenfunctions of a continuous-time and discrete-time linear time-invariant systems. Systems described by the differential equations and the difference equations. State-space analysis. Applications: Principle of Orthogonality, least squares problem, state space representation of continuous-time and discrete-time LTI systems. (9Hours)
- **Inner product spaces:** Determinants, Eigenvalues and Eigenvectors, Positive definite and semidefinite matrices, Inner product and L_p - norms; Singular Value Decomposition/ QR/ Schur & Principal Component Analysis, Pseudo-inverse of a full row/column-rank matrix, generalized inverse for a matrix which is non-singular and not a full full row/ column-rank. Applications: Best k-Rank approximation and matrix compression, fitting and linear regression problem; solving underdetermined/ overdetermined systems of linear equations. (12 Hours)
- **Optimization Methods:** Optimization framework description, linear and quadratic optimization problems, Lagrange dual function, geometric interpretation, and its applications; solving underdetermined/ overdetermined systems of linear equations with additional constraints like sparsity and regularization, etc. (12 Hours)

Laboratory Modules

- The two-hours of lab session per week will enhance the understanding of the concepts taught in the class. The lab will cover the concepts including principle of orthogonality, least Squares, SVD, low-rank matrix decompositions and its applications in the modern systems.

Text Books

1. Deisenroth, M. P., Faisal, A. A., Ong, C. S., **Mathematics for Machine Learning**, Cambridge University Press, 2020.
2. Strang, G., **Linear Algebra and Learning from Data**, Wellesley-Cambridge Press, 2019.
3. J. G. Proakis and M. Salehi, **Fundamentals of Communication Systems**, Prentice Hall, December 2004.

References

1. Friedberg, S.H., A Insel, A.J., A Spence, L.E., **Linear Algebra**, Pearson Education, 2014.
2. Boyd, S., & Vandenberghe, L., **Convex optimization**, Cambridge university press, 2004.
3. Alan V. Oppenheim, Ronald W. Schaffer., **Discrete-time signal processing**, Pearson Education, 2010.